

Tutorial

Know about your WiFicard

- Sudo lshw -C network

- ifconfig
- iwconfig

Manage connection in application layer

- Android => WPA_supplicant
- Linux => Network manager

WiFi interface modes

- `sudo iw list`

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Supported interface modes:
    * IBSS
    * managed
    * AP
    * AP/VLAN
    * WDS
    * monitor
    * mesh point
    * P2P-client
    * P2P-GO
software interface modes (can always be added):
    * AP/VLAN
    * monitor
```

Monitor mode

- airmon-ng
- iwconfig

System logs

- `tail -f /var/log/syslog`
- `dmesg` (Look for keywords - `wpa_supplicant`, `network manager`, `kernel`, `dhclient`)

Important and useful commands

- Wireless Events (scan, authentication etc) - `sudo iw event -t`
- Connection Stats (inactive time, rx/tx bytes/packets, signal, bitrate etc) - `sudo iw dev wlan0 station dump`
- Regular Monitoring - `watch -n1 "<command>"`
- `lshw -C network`, `lsusb`, `lspci`
- Debug Messages - `tail -f /var/log/syslog`, `dmesg`

Packet trace

- `sudo tcpdump -i <interface-name> -s <snaplength> <filters>`

- `sudo tcpdump -i eth0 -nn -s0 -v port 80`

- `sudo tcpdump -i eth0 dst 10.10.1.20 -w cap.pcap`

- **Tshark**

- `tshark -i wlan0 -w capture-output.pcap`

- `tshark -i wlan0 -Y http.request -T fields -e http.host -e http.user_agent`

Reference

- <https://www.aircrack-ng.org/doku.php?id=airmon-ng>
- <https://miloserdov.org/?p=126>
- https://www.wireshark.org/docs/wsug_html_chunked/ChCustCommandLine.html
- <https://www.tcpdump.org/manpages/tcpdump.1.html>